

scoup: Simulate Codon Sequences with Darwinian Selection Incorporated as an Ornstein-Uhlenbeck Process

Hassan Sadiq ^{1,2¶} and Darren P. Martin ²

¹ Department of Statistics and Actuarial Science, Stellenbosch University, South Africa ² Institute of Infectious Diseases and Molecular Medicine, Division of Computational Biology, Department of Integrative Biomedical Sciences, University of Cape Town, South Africa ¶ Corresponding author

DOI: [10.21105/joss.09797](https://doi.org/10.21105/joss.09797)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: Julia Romanowska 

Reviewers:

- [@clauswilke](#)
- [@ogoeli](#)

Submitted: 20 June 2025

Published: 05 March 2026

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](https://creativecommons.org/licenses/by/4.0/)).

Summary

Genetic analyses of natural selection within and between populations have increasingly developed along separate paths. The two important genres of evolutionary biology (i.e., phylogenetics and population genetics) borne from the split can only benefit from research that seeks to bridge the gap. Simulation algorithms that combine fundamental concepts from both genres are important to achieve such unifying objective. We introduce [scoup](#), a codon sequence simulator implemented in R and hosted on the Bioconductor platform. There is hardly any other simulator dedicated to genetic sequence generation for natural selection analyses on the platform. Concepts from the Halpern-Bruno mutation-selection model and the Ornstein-Uhlenbeck (OU) evolutionary algorithm were creatively fused such that the end-product is a novel simulator of genetic sequence evolution. Users are able to adjust the model parameters to mimic complex evolutionary procedures that may have been otherwise infeasible. For example, it is possible to explicitly interrogate the concepts of static and changing fitness landscapes with regards to Darwinian natural selection in the context of codon sequences from multiple populations.

Statement of need

Statistical inference models that are used to analyse the impact of Darwinian natural selection on observed genetic data, command a healthy portion of the phylogenetic literature ([Gupta & Vadde, 2023](#)). Validation of these largely codon-based models relies heavily on simulated data. Given the ever increasing diversity of natural selection inference models that exist ([Arenas, 2015](#); [Kosakovsky Pond et al., 2020](#); [Yang, 2007](#)), there is a need for more sophisticated simulators to match the expanding model complexities.

Bioconductor ([Gentleman et al., 2004](#)) is a leading bioinformatics platform distributing peer-reviewed R packages. A search of the entries on the platform, in Version 3.22 on 18 February 2026, with keywords including, codon, mutation, selection, simulate, and simulation returned a total of 70 packages (excluding [scoup](#)) out of the 2361 available. None of the retrieved entries was dedicated to codon data simulation for natural selection analyses. Thus, as an overdue contribution to the void, [scoup](#) is designed on the basis of the mutation-selection (MutSel) framework ([Halpern & Bruno, 1998](#)).

There are a few software packages simulating genetic sequences ([Peng et al., 2015](#)). Existing simulators tend to be more suitable for quantitative character evolution. These include, ape ([Paradis & Schliep, 2019](#)), ouch ([Butler & King, 2004](#); [Cressler et al., 2015](#)) and geiger ([Pennell et al., 2014](#)). Other extensively used DNA sequence simulators including, Seq-Gen

(Rambaut & Grass, 1997), INDELible (Fletcher & Yang, 2009), PhyloSim (Sipos et al., 2011) and phangorn (Schliep, 2011) are parameterized in accordance with ω -based models (Goldman & Yang, 1994; Muse & Gaut, 1994). More recent sequence simulators, such as, phastSim (Maio et al., 2022) and AliSim-HPC (Ly-Trong et al., 2023) prioritized output capacity. Only few genetic simulators were built upon the more elaborate MutSel evolutionary concept. These include, Pyvolve (Spielman & Wilke, 2015a) and SGWE (Arenas & Posada, 2014). To the best of our knowledge, these existing MutSel-friendly simulators are only able to generate data from static landscapes. With our proposed simulator, it is possible to generate codon sequences from landscapes that are static or those that are changing (also known as *seascapes*) (Mustonen & Lässig, 2010).

Algorithm

scoop is further unique for at least three reasons. First, it incorporates Darwinian natural selection into the MutSel model in terms of variability of selection coefficients, an extension of an idea from Spielman & Wilke (2015b). Second, it directly utilises the concept of fitness landscapes (Wright, 1932). Third, fitness landscape updates can be executed in either a deterministic or a stochastic format. The stochastic updates are implemented in terms of the more biologically amenable, Ornstein-Uhlenbeck (OU) process (Bartoszek et al., 2017; Uhlenbeck & Ornstein, 1930). A crude summary of how substitution events are executed in **scoop** is presented in Figure 1.

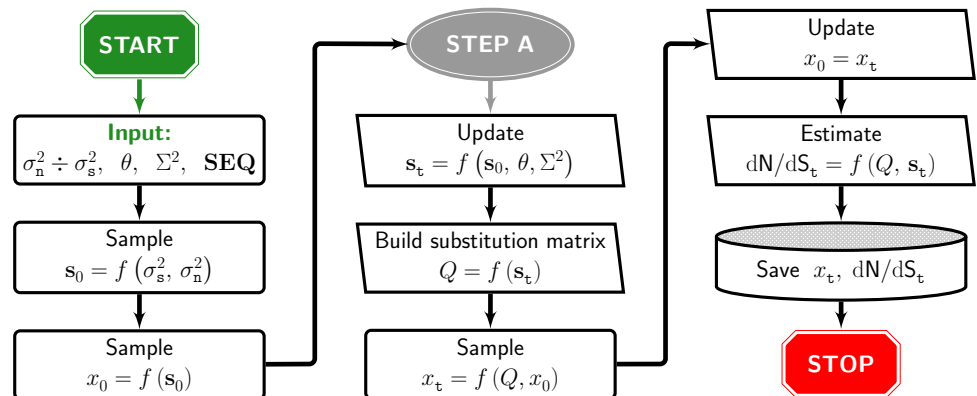


Figure 1: Summarised **scoop algorithm.** The flowchart shows the process for a single substitution event. After each substitution event, the process returns to *STEP A*, until the input tree length ($\tau \in \text{SEQ}$) is exhausted. σ_n^2 = variance of amino acid selection coefficients. σ_s^2 = variance of synonymous codon selection coefficients. Σ^2 = OU asymptotic variance. θ = OU mean reversion rate. **SEQ** = sequence information. x_* = codon. s_* = codon selection coefficient vector.

We highlight two important design choices from Figure 1. First, we assume that a static fitness landscape is obtained from a single set of parameters (ξ) needed to sample a 20-element numerical vector of amino acid selection coefficients (that is, s_0 in Figure 1). The coefficients are subsequently used as inputs of the corresponding MutSel model. The seascape setting is then defined as a function of multiple sets of parameters ($\xi_1, \xi_2, \dots, \xi_k$, for $k \leq \text{extant taxa size}$). Second, the coefficient update (s_t) step is done after every substitution event. In addition, the Ornstein-Uhlenbeck update process is discretised. In other words, the OU jump sizes are fixed and pre-specified as an input to the simulation functions.

Exemplary results

`scoup` is primarily designed using base functions in R. Some important complementary functions are imported from the Matrix (Bates et al., 2024) and the Biostrings (Pagès et al., 2024) packages. We simulated some sequences with `scoup` to verify the accuracy of the outputs from the package. The output data comprise eight sequences and 1000 codon sites. All the other necessary model parameters were kept the same for all simulated replicates. The data and all the associated files, including the simulation and analyses code, are available in the paper/data folder as part of the package. Only the variance of the selection coefficients of the synonymous codons ($\sigma_s^2 = \{0.00, 0.01\}$) and the variance of the amino acids ($\sigma_n^2 = \{0.001, 0.005, 0.009, 0.030, 0.070, 0.100, 0.500, 0.900\}$) were varied, with five replicate sequences generated for each (σ_s^2, σ_n^2) combination. The data sets were analysed with PAML (Yang, 2007) to obtain maximum likelihood estimates of the ratio of the rates of non-synonymous to synonymous substitutions (ω) and these were compared to the analytical estimates (dN/dS) obtained from `scoup`. The results are summarised in Figure 2.

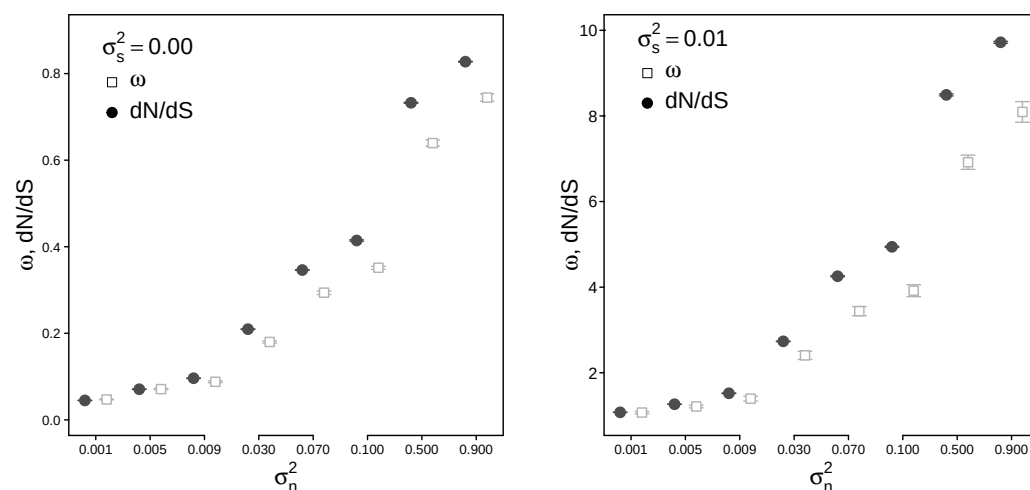


Figure 2: Analyses of data generated with `scoup`. The width of each arrow is proportional to the standard error. σ_n^2 = variance of amino acid selection coefficients. σ_s^2 = variance of synonymous codon selection coefficients. ω = maximum likelihood estimate of non-synonymous to synonymous substitution rates ratio obtained using PAML, dN/dS = analytical equivalent of ω that is returned as part of the outputs from `scoup`.

Three features from Figure 2 are noteworthy. First, there is good correlation between the simulated (as measured by dN/dS) and the inferred (as measured by ω) magnitude of natural selection effect. The slight discrepancies as σ_n^2 increases are likely due to the limited sizes of the data sets (Spielman & Wilke, 2015a). This implies that outputs from `scoup` are reliable. Second, as expected (Spielman & Wilke, 2015b), in the absence of synonymous selection (that is, $\sigma_s^2 = 0$) the selection effect is predominantly negative (that is, $dN/dS, \omega < 1$) and the effect is largely positive when synonymous selection is present. This further asserts of the reliability of the outputs from `scoup`. Third, the magnitude of natural selection effect may be influenced by amino acid selection (or aptly, non-synonymous selection). This property is yet to be thoroughly investigated in the computational molecular evolution literature and there is hardly any other available computational resource that permits its exploration. This underlines the potential importance of `scoup`.

Conclusions

We present [scoup](#), an R package for codon sequences simulation, where molecular evolutionary processes are mirrored more realistically than most existing simulators. Our framework creatively incorporates the Ornstein-Uhlenbeck process into the mutation-selection evolutionary model. This attribute could potentially unlock exciting research avenues that will improve existing knowledge about the complex interactions of different, potentially interacting, molecular evolutionary processes.

Code availability

[scoup](#) is published for free public use under the GPL-2 license. It is available for download from the [Bioconductor platform](#), along with detailed documentation and tutorial files. Some additional sample code are accessible in the [tests](#) and the [vignettes](#) folders of the package.

Acknowledgements

We thank Ben Murrell for suggesting modelling varying selection coefficients with an OU process. Computations were performed using the [HPC](#) facility at Stellenbosch University, South Africa.

References

- Arenas, M. (2015). Trends in Substitution Models of Molecular Evolution. *Frontiers in Genetics*, 6(319). <https://doi.org/10.3389/fgene.2015.00319>
- Arenas, M., & Posada, D. (2014). Simulation of Genome-wide Evolution under Heterogeneous Substitution Models and Complex Multispecies Coalescent Histories. *Molecular Biology and Evolution*, 31(5), 1295–1301. <https://doi.org/10.1093/molbev/msu078>
- Bartoszek, K., Glémin, S., Kaj, I., & Lascoux, M. (2017). Using the Ornstein–Uhlenbeck Process to Model the Evolution of Interacting Populations. *Journal of Theoretical Biology*, 429, 35–45. <https://doi.org/10.1016/j.jtbi.2017.06.011>
- Bates, D., Maechler, M., & Jagan, M. (2024). *Matrix: Sparse and Dense Matrix Classes and Methods*. <https://doi.org/10.32614/CRAN.package.Matrix>
- Butler, M. A., & King, A. A. (2004). Phylogenetic Comparative Analysis: A Modeling Approach for Adaptive Evolution. *The American Naturalist*, 164(6), 683–695. <https://doi.org/10.1086/426002>
- Cressler, C. E., Butler, M. A., & King, A. A. (2015). Detecting Adaptive Evolution in Phylogenetic Comparative Analysis Using the Ornstein-Uhlenbeck Model. *Systematic Biology*, 64(6), 953–968. <https://doi.org/10.1093/sysbio/syv043>
- Fletcher, W., & Yang, Z. (2009). INDELible: A Flexible Simulator of Biological Sequence Evolution. *Molecular Biology and Evolution*, 26(8), 1879–1888. <https://doi.org/10.1093/molbev/msp098>
- Gentleman, R. C., Carey, V. J., Bates, D. M., Bolstad, B., Dettling, M., Dudoit, S., Ellis, B., Gautier, L., Ge, Y., Gentry, J., Hornik, K., Hothorn, T., Huber, W., Iacus, S., Irizarry, R., Leisch, F., Li, C., Maechler, M., Rossini, A. J., ... Zhang, J. (2004). Bioconductor: Open Software Development for Computational Biology and Bioinformatics. *Genome Biology*, 5(10), R80. <https://doi.org/10.1186/gb-2004-5-10-r80>
- Goldman, N., & Yang, Z. (1994). A Codon-based Model of Nucleotide Substitution for

- Protein-coding DNA Sequences. *Molecular Biology and Evolution*, 11(5), 725–736. <https://doi.org/10.1093/oxfordjournals.molbev.a040153>
- Gupta, M. K., & Vadde, R. (2023). Next-Generation Development and Application of Codon Model in Evolution. *Frontiers in Genetics*, 14. <https://doi.org/10.3389/fgene.2023.1091575>
- Halpern, A. L., & Bruno, W. J. (1998). Evolutionary Distances for Protein-Coding Sequences: Modelling Site-Specific Residue Frequencies. *Molecular Biology and Evolution*, 15(7), 910–917. <https://doi.org/10.1093/oxfordjournals.molbev.a025995>
- Kosakovsky Pond, S. L., Poon, A. F. Y., Velazquez, R., Weaver, S., Hepler, N. L., Murrell, B., Shank, S. D., Magalis, B. R., Bouvier, D., Nekrutenko, A., Wisotsky, S., Spielman, S. J., Frost, S. D. W., & Muse, S. V. (2020). HyPhy 2.5 – A Customizable Platform for Evolutionary Hypothesis Testing Using Phylogenies. *Molecular Biology and Evolution*, 37(1), 295–299. <https://doi.org/10.1093/molbev/msz197>
- Ly-Trong, N., Barca, G. M. J., & Minh, B. Q. (2023). AliSim-HPC: Parallel Sequence Simulator for Phylogenetics. *Bioinformatics*, 39(9), btad540. <https://doi.org/10.1093/bioinformatics/btad540>
- Maio, N. D., Boulton, W., Weilguny, L., Walker, C. R., Turakhia, Y., Corbett-Detig, R., & Goldman, N. (2022). phastSim: Efficient Simulation of Sequence Evolution for Pandemic-Scale Datasets. *PLOS Computational Biology*, 18(4), e1010056. <https://doi.org/10.1371/journal.pcbi.1010056>
- Muse, S. V., & Gaut, B. S. (1994). A Likelihood Approach for Comparing Synonymous and Nonsynonymous Nucleotide Substitution Rates, with Application to the Chloroplast Genome. *Molecular Biology and Evolution*, 11(5), 715–724. <https://doi.org/10.1093/oxfordjournals.molbev.a040152>
- Mustonen, V., & Lässig, M. (2010). Fitness Flux and Ubiquity of Adaptive Evolution. *Proceedings of the National Academy of Sciences*, 107(9), 4248–4253. <https://doi.org/10.1073/pnas.0907953107>
- Pagès, H., Aboyoun, P., Gentleman, R., & DebRoy, S. (2024). *Biostrings: Efficient Manipulation of Biological Strings*. <https://doi.org/10.18129/B9.bioc.Biostrings>
- Paradis, E., & Schliep, K. (2019). ape 5.0: An Environment for Modern Phylogenetics and Evolutionary Analyses in R. *Bioinformatics*, 35(3), 526–528. <https://doi.org/10.1093/bioinformatics/bty633>
- Peng, B., Chen, H.-S., Mechanic, L. E., Racine, B., Clarke, J., Gillanders, E., & Feuer, E. J. (2015). Genetic Data Simulators and their Applications: An Overview. *Genetic Epidemiology*, 39(1), 2–10. <https://doi.org/10.1002/gepi.21876>
- Pennell, M. W., Eastman, J. M., Slater, G. J., Brown, J. W., Uyeda, J. C., FitzJohn, R. G., Alfaro, M. E., & Harmon, L. J. (2014). geiger v2.0: An Expanded Suite of Methods for Fitting Macroevolutionary Models to Phylogenetic Trees. *Bioinformatics*, 30(15), 2216–2218. <https://doi.org/10.1093/bioinformatics/btu181>
- Rambaut, A., & Grass, N. C. (1997). Seq-Gen: An Application for the Monte Carlo Simulation of DNA Sequence Evolution Along Phylogenetic Trees. *Bioinformatics*, 13(3), 235–238. <https://doi.org/10.1093/bioinformatics/13.3.235>
- Schliep, K. P. (2011). phangorn: Phylogenetic Analysis in R. *Bioinformatics*, 27(4), 592–593. <https://doi.org/10.1093/bioinformatics/btq706>
- Sipos, B., Massingham, T., Jordan, G. E., & Goldman, N. (2011). PhyloSim – Monte Carlo Simulation of Sequence Evolution in the R Statistical Computing Environment. *BMC Bioinformatics*, 12(104). <https://doi.org/10.1186/1471-2105-12-104>
- Spielman, S. J., & Wilke, C. O. (2015a). Pyvolve: A Flexible Python Module for Simulating

- Sequences along Phylogenies. *PLoS ONE*, 10(9), e0139047. <https://doi.org/10.1371/journal.pone.0139047>
- Spielman, S. J., & Wilke, C. O. (2015b). The Relationship between dN/dS and Scaled Selection Coefficients. *Molecular Biology and Evolution*, 32(4), 1097–1108. <https://doi.org/10.1093/molbev/msv003>
- Uhlenbeck, G. E., & Ornstein, L. S. (1930). On the Theory of the Brownian Motion. *Physical Review*, 36, 823–841. <https://doi.org/10.1103/PhysRev.36.823>
- Wright, S. (1932). The Roles of Mutation, Inbreeding, Crossbreeding and Selection in Evolution. *Proceedings of the Sixth International Congress of Genetics*, 1, 356–366.
- Yang, Z. (2007). PAML 4: Phylogenetic Analysis by Maximum Likelihood. *Molecular Biology and Evolution*, 24(8), 1586–1591. <https://doi.org/10.1093/molbev/msm088>